

The Next Generation of ReShare

Architecture, adaptable workflows, interoperability, and migration

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From CrossLink broker to CrossLink platform

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WOLFcon 2025	WOLFcon 2026
Standards-based ILL broker	Complete ILL workflow platform
Connect external peers	Native borrowing and lending
Route and observe transactions	Model and operate the lifecycle

New: Patron Requests API · NCIP · explicit state model · live UI events · scheduling

What ReShare proved

- Community-owned resource sharing works
- ILS-neutral workflows can operate in production
- Practitioners can shape the service
- Open standards connect diverse systems

What needed to change

- Substantial application and persistence framework overhead
- Tenant topology multiplied operational work
- Kafka added infrastructure and failure windows
- Workflow behavior was spread across the application
- Changes required tracing, releases, and broad testing

We are replacing an implementation, not the product.

Preserve

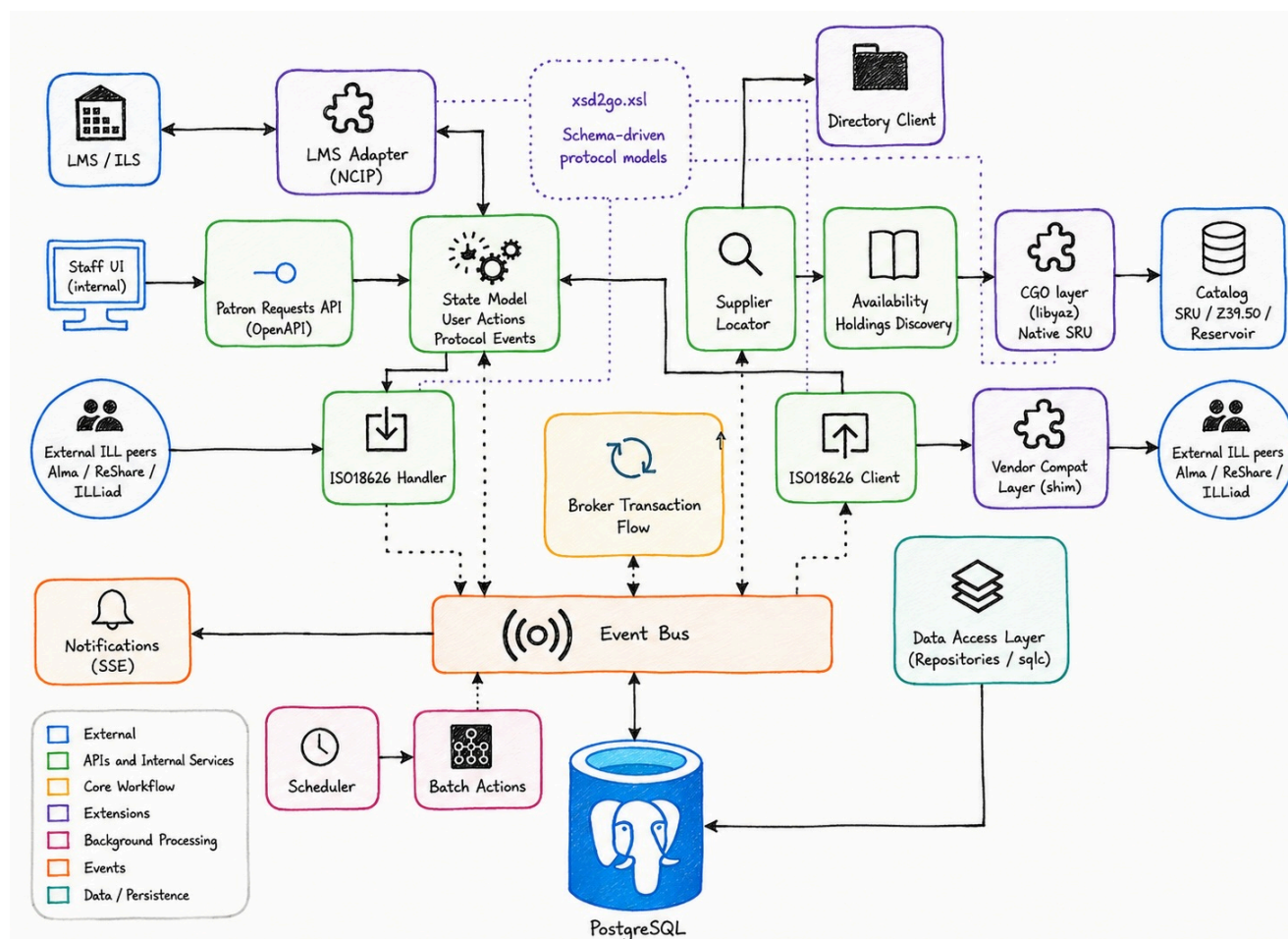
- ILS/LSP neutrality
- First-class ISO 18626 and NCIP
- Borrowing and lending workflows
- Consortial routing and supplier selection
- Multi-tenant operation

Improve

- Smaller, more reliable operational footprint
- Explicit and adaptable workflows
- Observable and auditable requests
- One shared deployment for virtual tenants
- Support for native direct borrowing across the consortium
- A richer, OpenAPI-based Directory for networks, tiers, policies, and integrations (NCIP, Z39.50)

A platform built around the transaction

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Open protocols at the edges · explicit workflow at the center · durable work underneath

Patron Request

- The lifecycle practitioners operate
- Borrowing or lending perspective
- Current state and available actions
- Items, notifications, and attention flags
- Search, filtering, and batch work

ILL Transaction

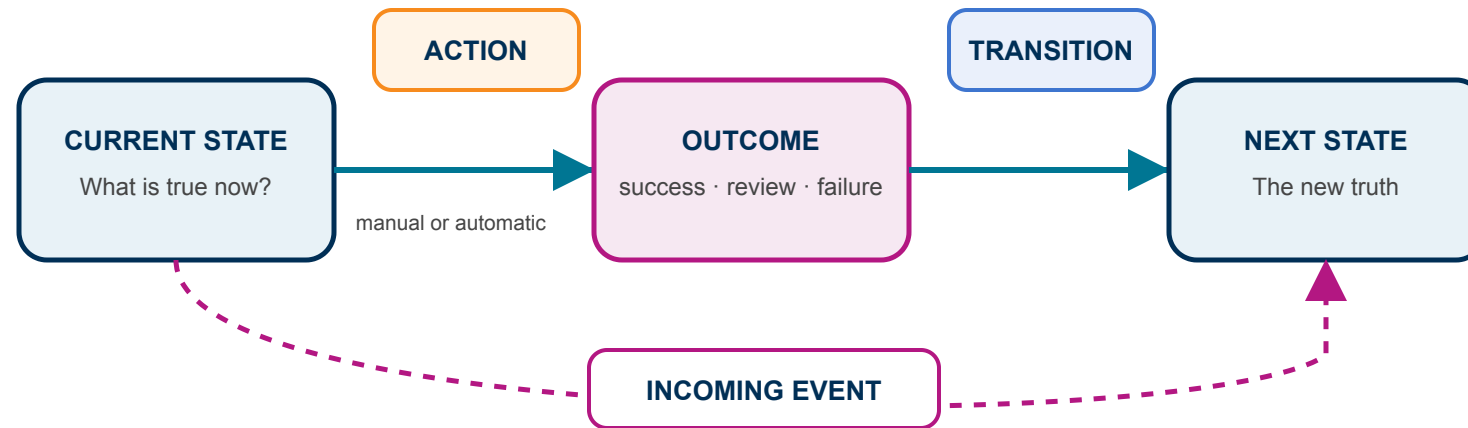
- Supplier and rota decisions
- ISO 18626 message exchange
- Adapter and protocol activity
- Complete durable event history

A hyperlinked API connects the request, transaction, protocol activity, and event history—from intent to outcome.

Workflow is now a YAML model

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A small grammar for a complete workflow

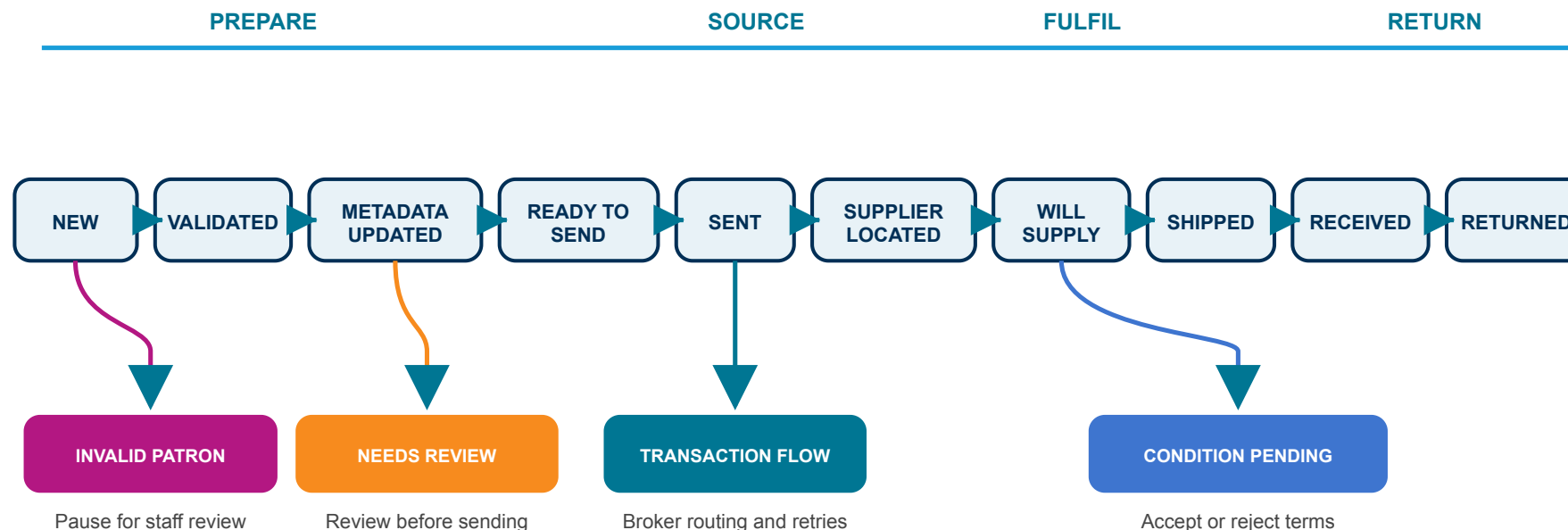


External messages use the same permitted transitions

```
- name: NEW
  side: REQUESTER
  initial: true
  primaryAction: validate-patron
  actions:
    - name: validate-patron
      trigger: auto
      transitions: {success: VALIDATED, review: INVALID_PATRON}
```

A borrowing request, step by step

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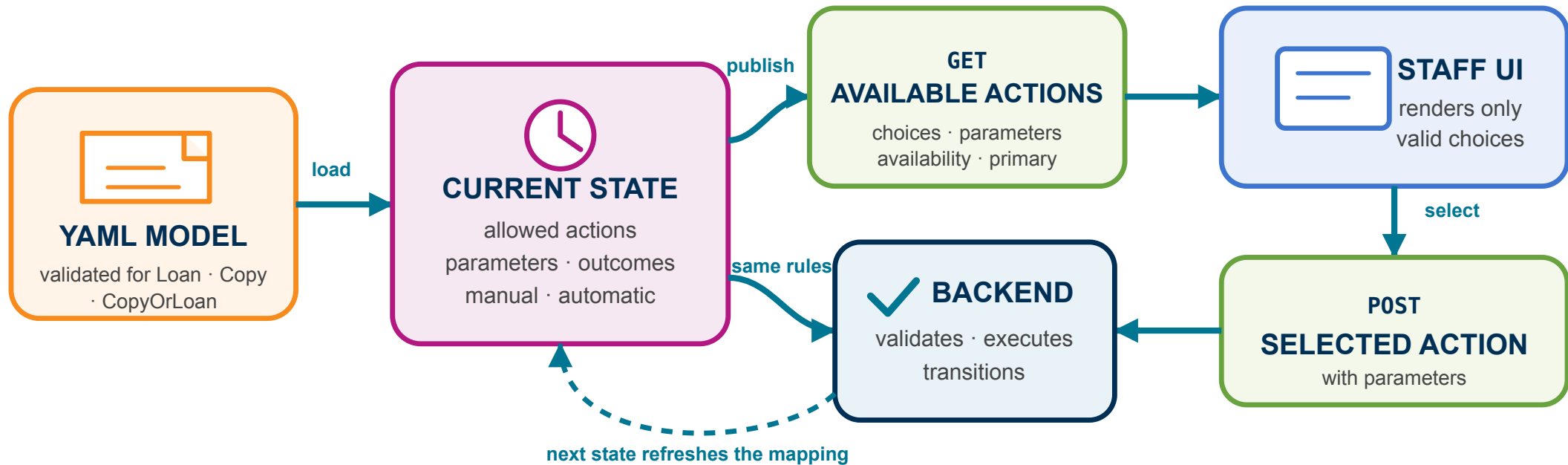


The happy path is simple. The model makes exceptions explicit.

The happy path and every exception are explicit, inspectable, and testable.

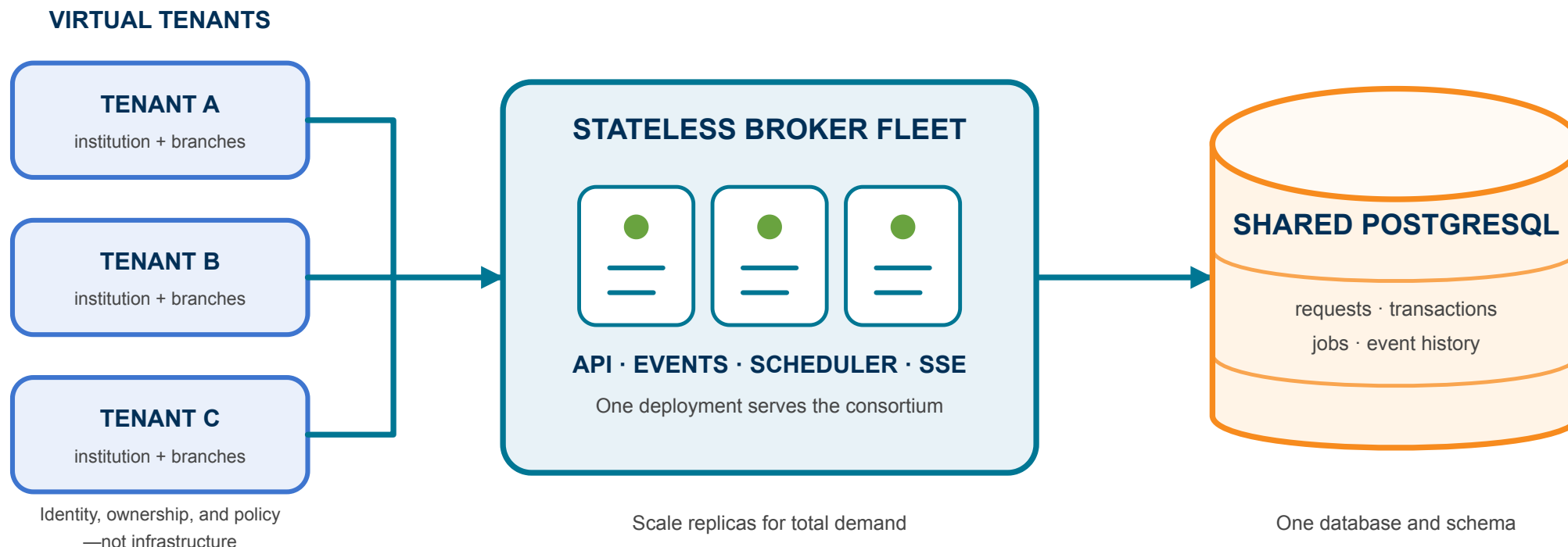
One workflow model drives the API and UI

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One deployment, many virtual tenants

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The new architecture enables native DCB support in ReShare

Union catalog → live availability → internal fulfillment → local ILS

A shared deployment is an option, not a constraint

Independent local brokers can still form a peer-to-peer ILL network

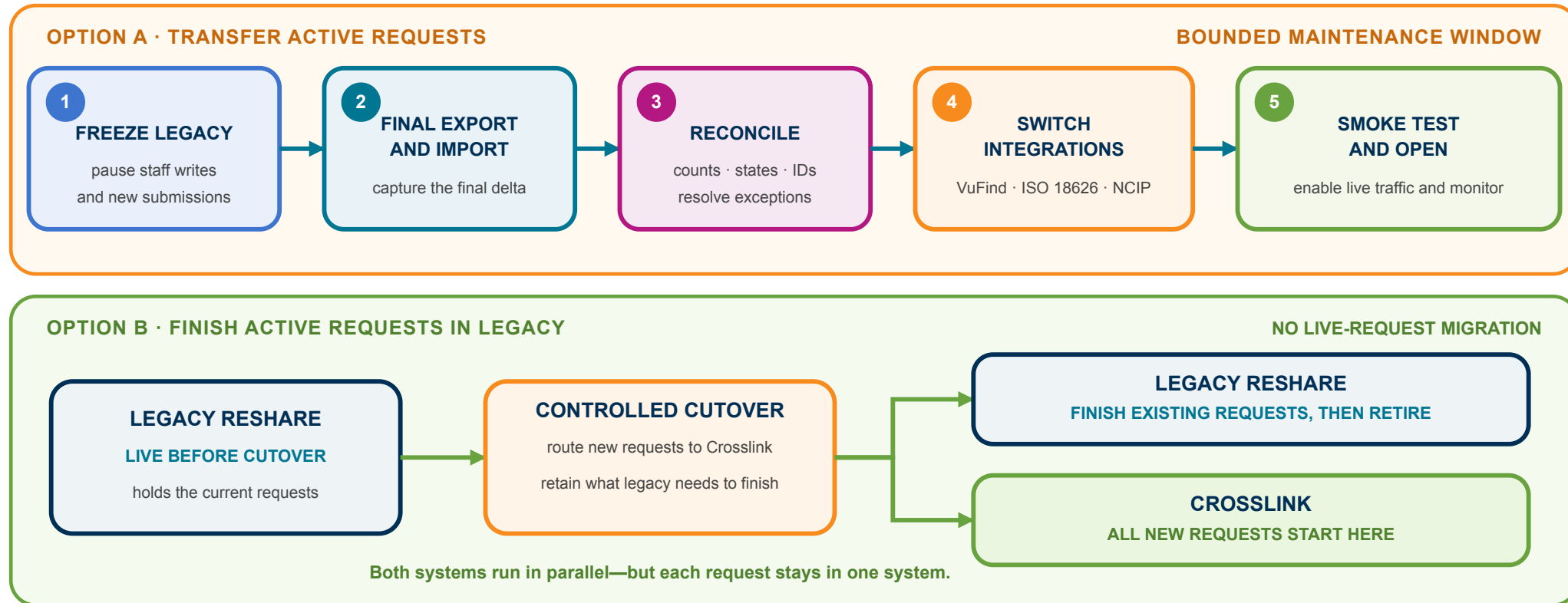
Measure	New broker	mod-rs 2.13	Difference
Container image	35.6 MB	246.0 MB	6.9× smaller
Slowest cold start—3 runs	0.667 s	37.006 s	55× faster
App working set (RSS)	19.7 MiB	1,509.2 MiB	77× lower
Core runtime pieces	2	5	3 fewer
Test-to-production LOC ratio	1.71:1	0.16:1	10.9× higher

Native ARM and a 2 GiB app limit. Legacy app tier includes Kafka + ZooKeeper; PostgreSQL and Okapi excluded.

A smaller operational footprint means simpler deployment, recovery, scaling, and upgrades.

Adoption with a migration plan

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Rehearse with real data, prove readiness, and choose the cutover path that fits each consortium.